



2025

D24(B)

BOOKLET NO.

249081

Zoology

Paper – II

Time Allowed : Three Hours

Maximum Marks : 250

Medium : English

Type of Paper : Conventional /Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Keep in mind the word limit indicated in the question if any.
8. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
9. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
10. Neat sketches may be drawn, wherever required.

P.T.O.

**SECTION – A**

Q1. Answer the following questions in about **150** words : **(10×5=50)**

Q1. (a) Write on eukaryotic ribosome. **10**

Q1. (b) Describe the characteristics of Genetic code. **10**

Q1. (c) Describe the evolution of horse. **10**

Q1. (d) Explain the principles of zoological nomenclature. **10**

Q1. (e) Describe the structure of lampbrush chromosome. **10**

Q2. (a) Give an account of ultrastructure of mitochondria. Also discuss the role of mitochondria in ATP formation. **20**

Q2. (b) Write principle and applications of RAPD. **15**

Q2. (c) Explain the phenomenon of natural selection taking examples of sickle cell anemia and peppered moth. **15**

Q3. (a) Define mutation. Explain the causes and types of mutation. **20**

Q3. (b) Give an account of RNA polymerases and promoters involved in eukaryotic transcription. **15**

Q3. (c) What is Hardy-Weinberg Law ? How is gene frequency changed by mutation and genetic drift in populations ? **15**

Q4. (a) Explain the continental drift. How does the continental drift theory explain the discontinuous and restricted distribution of certain animal species ? **20**

Q4. (b) Define Meiosis. Explain the Prophase-I of Meiosis-I. **15**

Q4. (c) Comment on Human genome project. **15**

**SECTION – B**

- Q5.** Answer the following questions in about **150** words : **(10×5=50)**
- Q5. (a)** Describe the different steps involved in Kreb's cycle. **10**
- Q5. (b)** Comment on source of energy during muscle contraction under normal and intensive exercise condition. **10**
- Q5. (c)** Comment on stem cells. **10**
- Q5. (d)** Describe the complete pathway of Glycolysis. **10**
- Q5. (e)** Discuss the structure and functions of cyclic AMP in cellular metabolism. **10**
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- Q6. (a)** Describe the process of spermatogenesis. **20**
- Q6. (b)** Explain the sources and functions of Vitamin B₁, B₂ and B₁₂. **15**
- Q6. (c)** Explain the structure of Haemoglobin and it's role in carbon dioxide transport. **15**
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- Q7. (a)** Explain the structure of nephron. Add a note on hormonal regulation of urine formation. **20**
- Q7. (b)** Describe the structure, types and functions of immunoglobulins. **15**
- Q7. (c)** Describe in detail three phases of oogenesis in Human Females. **15**
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- Q8. (a)** Explain enzymes and their characteristics as well as lock and key model of mechanism of enzyme action. **20**
- Q8. (b)** Describe the role of salivary gland and pancreas in digestion and absorption process in mammals. **15**
- Q8. (c)** Describe the organogenesis of an eye. **15**
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